



Education

University of California, Riverside

June 2019

- B.S. in Computer Science | GPA: 3.4 / 4.0
- Relevant Courses: Computer Graphics, Virtual Reality, Software Engineering, Operating Systems, Linear Algebra, Multivariable Calculus, Physics, Data Structures & Algorithms, Machine Learning

Skills

- C++, C#, Python, Git, OpenGL, Unity 3D, Matlab, Android SDK, Maya, Photoshop, LaTeX, Agile Methodologies, Scrum, Mac OSX, Windows, Linux, Vim, Computer Graphics, Virtual Reality

Professional Experience

Computer Science Part-Time Tutor. The Coding School. Independent

Sept. 2019 - Present

- Teach computer science concepts using Python3 to middle school students via Skype
- Establish the Android development curriculum by coding a drum sample media player in Java

Visiting Researcher Intern: Real-time Rendering for Virtual Humans

June 2018 - Aug. 2018

University of Southern California Institute for Creative Technologies. Playa Vista, CA

- Developed the blend-shape (animation) pipeline in C# to view and animate facial rigs in Unity 3D
- Coded a graphical user interface using python scripts to animate facial scans
- Parsed through data files efficiently in C++ containing vertices of 3 facial expressions

Computer Graphics Assistant. UC Riverside. Riverside, CA

Jan. 2018 - Feb. 2018

- Programmed POV-Ray test scripts to render OpenGL and PhysBAM simulation examples
- Collected and compiled photo stills using Linux/Unix commands to produce movie files
- Organized a Kanban board to communicate status of renderings

Technical Projects

VR Vapors. Google Cardboard Virtual Reality Developer

July 2019 - Aug. 2019

- Coded wave and particle visualizations based on song frequencies in C# through Unity

LED Strip: Visualizing Sorting Algorithms. Embedded Systems - [VIDEO](#)

Apr. 2019 - June 2019

- Programmed in C an individually addressable LED strip to display two sorting algorithms
- Synchronized data transfers between microcontroller and Arduino with USART

Ball Hero. Unity 3D Game Developer - [GITHUB](#)

Apr. 2019 - June 2019

- Led a team of 5 in creating a 3D rolling ball game using a scrum methodology
- Developed in-game enemies with C# scripts

Pómo. Independent Study: Technical Director

Jan. 2019 - March 2019

- Illustrated storyline and concept art for assets and characters of an original animated short
- Wrote Maya and Python scripts for character rigging and animation

Personal Website

Dec. 2018 - Jan. 2019

- Created a portfolio website with Bootstrap framework while editing HTML, CSS, and Javascript code
- Incorporated videos into webpage through HTML5 Video
- Tracked user trends using Google Analytics

Carnival Games. Virtual Reality Game Developer - [GITHUB](#)

Sept. 2018 - Dec. 2018

- Collaborated with a team of 3 to assemble the game logic in C# and Unity for Oculus Rift
- Designed 3D Ball Toss game assets in Maya and Unity
- Maximized an immersive gaming experience by including sound effects and visual simulation

Parking App. Android Developer - [VIDEO](#)

Sept. 2018 - Dec. 2018

- Built an Android app in Java that assists students in finding available parking at UC Riverside
- Generated parking trends from university servers using data stored in Google's Firebase
- Directed a team of 4 in user interface design choices as scrum master

MiniGL Pipeline Developer. C++ Developer - [GITHUB](#)

Sept. 2017 - Dec. 2017

- Built a lite version of the OpenGL graphics pipeline in C++
- Implemented matrix transformations, rasterization, and ray tracing techniques

Volunteer Experience

- SIGGRAPH Student Volunteer. Los Angeles, CA

July 2019 - Aug. 2019